

Junkai Zhang

CS PhD Candidate @ UCLA

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Education

University of California, Los Angeles

Sep. 2022 – Jun. 2027 (Expected)

Ph.D. in Computer Science, advisor: Professor Wei Wang / Los Angeles, CA, US

Tsinghua University

Aug. 2018–Jun. 2022

B.S. in Mathematics, Minor in Philosophy / Beijing, CN

Research Interests

LLM Post-training, LLM Agents

Professional Experience

Google

Oct. 2025 – Current

Student Researcher / Playa Vista, CA, US

Scale AI

Jun. 2025 – Sep. 2025

Post-training Research Intern / San Francisco, CA, US

Selected Works

The asterisk * indicates equal contributions.

Chasing the Tail: Effective Rubric-based Reward Modeling for Large Language Model Post-Training

Junkai Zhang*, Zihao Wang*, Lin Gui*, Swarnashree Mysore Sathyendra, Jaehwan Jeong, Victor Veitch, Wei Wang, Yunzhong He, Bing Liu, Lifeng Jin

in *ICLR 2026*

DuoGuard: A Two-Player RL-Driven Framework for Multilingual LLM Guardrails

Yihe Deng*, Yu Yang*, **Junkai Zhang***, Wei Wang, Bo Li

in *AISTATS 2026*

Publications and Manuscripts

The asterisk * indicates equal contributions.

REINFORCEMENT LEARNING

Uncertainty-Aware Reward-Free Exploration with General Function Approximation

Junkai Zhang*, Weitong Zhang*, Dongruo Zhou, Quanquan Gu

in *ICML 2024*

Optimal Horizon-Free Reward-Free Exploration for Linear Mixture MDPs

Junkai Zhang, Weitong Zhang, Quanquan Gu

in *ICML 2023*

AI FOR SCIENCE —————

MatSciBench: Benchmarking the Reasoning Ability of LLM in Material Science

Junkai Zhang*, Jingru Gan*, Zian Jia, Changquan Gu, Jianpeng Chen, Xiaoxuan Wang, Yanqiao Zhu, Mingyu Derek Ma, Dawei Zhou, Ling Li, Wei Wang

in *KDD 2026 AI for Sciences Track*

MetaScientist: A Human-AI Synergistic Framework for Automated Mechanical Metamaterial Design

Jingyuan Qi, Zian Jia, Minqian Liu, Wangzhi Zhan, **Junkai Zhang**, Xiaofei Wen, Jingru Gan, Jianpeng Chen, Qin Liu, Mingyu Derek Ma, Bangzheng Li, Haohui Wang, Adithya Kulkarni, Muhao Chen, Dawei Zhou, Ling Li, Wei Wang, Lifu Huang

in *NAACL 2025 - Demo Track*

Protein Large Language Models: A Comprehensive Survey

Yijia Xiao, Wanjia Zhao, **Junkai Zhang**, Yiqiao Jin, Han Zhang, Zhicheng Ren, Renliang Sun, Haixin Wang, Guancheng Wan, Pan Lu, Xiao Luo, Yu Zhang, James Zou, Yizhou Sun, Wei Wang

in *EMNLP 2025 Findings*

MetamatBench: Integrating Heterogeneous Data, Computational Tools, and Visual Interface for Metamaterial Discovery

Jianpeng Chen, Wangzhi Zhan, Haohui Wang, Zian Jia, Jingru Gan, **Junkai Zhang**, Jingyuan Qi, Tingwei Chen, Lifu Huang, Muhao Chen, Ling Li, Wei Wang, Dawei Zhou

in *KDD 2025 Datasets and Benchmarks Track*

Neural network-assisted personalized handwriting analysis for Parkinson’s disease diagnostics

Guorui Chen, Trinny Tat, Yihao Zhou, Zhaoqi Duan, **Junkai Zhang**, Kamryn Scott, Xun Zhao, Zeyang Liu, Wei Wang, Song Li, Katy A. Cross, Jun Chen

in *Nature Chemical Engineering*

Continuous Treatment Effect Modeling in Multi-agent Dynamical Systems

Zijie Huang*, Jeehyun Hwang*, **Junkai Zhang***, Jinwoo Baik, Weitong Zhang, Quanquan Gu, Dominik Wodarz, Yizhou Sun, Wei Wang

in *WWW 2024*

MACHINE LEARNING —————

Fast Sampling via De-randomization for Discrete Diffusion Models

Zixiang Chen, Huizhuo Yuan, Yongqian Li, Yiwen Kou, **Junkai Zhang**, Quanquan Gu

in *NeurIPS 2024*

Why Does Sharpness-Aware Minimization Generalize Better Than SGD?

Zixiang Chen*, **Junkai Zhang***, Yiwen Kou, Xiangning Chen, Cho-Jui Hsieh, Quanquan Gu

in *NeurIPS 2023*

Academic Services

Conference Reviewer: ICML[2025], ICLR[2024-2026], NeurIPS[2025], NAACL[2025], AAAI[2024-2025], AISTATS[2024]